

Examiner
only

2. The table provides information about three animals.

	Annelid (e.g. earthworm)	Amphibian (e.g. adult frog)	Mammal (e.g. human)
Environmental conditions	Terrestrial (moist)	Terrestrial (moist)	Terrestrial
Gas exchange surface	Skin	Skin, buccal cavity and lungs	Lungs
Permeability of skin to gases	Permeable	Permeable	Impermeable
Mean skin surface area /mm ²		5 × 10 ⁴	1.8 × 10 ⁶
Mean body volume /mm ³	1.2 × 10 ³	2.5 × 10 ⁴	6 × 10 ⁷
Mean surface area to volume ratio	3 : 1	2 : 1	0.03 : 1

(a) Calculate the mean skin surface area of the annelid. **Give your answer in standard form.** [2]

Surface area = mm²

(b) The animals shown in the table have closed circulatory systems. Describe what is meant by a closed circulatory system and explain its role in these animals. [2]

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(c) The amphibian and the mammal possess mechanisms to ventilate their lungs.

Use the information in the table and your own knowledge, to explain why the amphibian does not need to ventilate its lungs when inactive, whereas the mammal must ventilate its lungs all the time. [4]

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For a given species, the stomatal number may vary considerably but the stomatal index is relatively constant. Explain this observation with reference to **Table 1** and **Diagrams 1** and **2**. [2]

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